Sulyab Thottungal Valapu

Contact Information

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Research Interests

Internet Measurement, DNS Security and Resilience, IPv6

Education

University of Southern California
Ph.D. Computer Science, 2021 – 2027 (Expected)
Thesis: Characterizing the use of Internet naming and addressing to improve security and deployment

Indian Institute of Information Technology, Allahabad
B.Tech. Information Technology, 2014 – 2018

Industry & Research Experience

PhD Research Intern
Verisign, 2026/05 – 2026/08
Evaluating the security of domain tokenization platforms

Software Engineer
Arista Networks, 2018/07 – 2020/12
Development and maintenance of network layer software for Arista EOS

Software Engineering Intern
Arista Networks, 2018/01 – 2018/06
Improving debuggability of OSPF and OSPFv3 implementations

Research Intern
Yonsei University, 2017/05 – 2017/07
Evaluating the use of blockchain for consensus among intelligent vehicles

Publications

1. "Towards a Non-Binary View of IPv6 Adoption." Sulyab Thottungal Valapu and John Heidemann. Proceedings of the 2025 ACM Internet Measurement Conference (IMC), New York, NY, USA, pp. 727–745, November 2025. DOI: 10.1145/3730567.3764467.
2. "Geofeed Adoption and Authentication." Dipsy Desai, Kicho Yu, and Sulyab Thottungal Valapu. 2025 IEEE Network Operations and Management Symposium (NOMS), pp. 01–08, May 2025. DOI: 10.1109/NOMS57970.2025.11073659.
3. "Reward-Based Blockchain Infrastructure for 3D IC Supply Chain Provenance." Sulyab Thottungal Valapu, Aritri Saha, Bhaskar Krishnamachari, Vivek Menon, and Ujjwal Guin. 2025 IEEE International Symposium on Hardware Oriented Security and Trust (HOST), pp. 370–380, May 2025. DOI: 10.1109/HOST64725.2025.11050057.
4. "DARSAN: A Decentralized Review System Suitable for NFT Marketplaces." Sulyab Thottungal Valapu, Tamoghna Sarkar, Jared Coleman, Anusha Avyukt, Hugo Embrechts, Dimitri Torfs, Michele Minelli, and Bhaskar Krishnamachari. Blockchain – ICBC 2023, Cham, pp. 3–20, 2023. DOI: 10.1007/978-3-031-44920-8_1.

Posters

1. "Crowd-SFT: Crowdsourcing for LLM Alignment." Alex Sotiropoulos, Sulyab Thottungal Valapu, Linus Lei, Jared Coleman, and Bhaskar Krishnamachari. 2025 IEEE International Conference on Decentralized Applications and Infrastructures (DAPPS), pp. 135–136, July 2025. DOI: 10.1109/DAPPS65174.2025.00026.

Invited Talks

1. "Towards a Non-Binary View of IPv6 Adoption." NANOG 97, Bellevue, WA, USA, June 2026.

Technical Reports

1. "Zombies in Alternate Realities: The Afterlife of Domain Names in DNS Integrations." Sulyab Thottungal Valapu, John Heidemann, Mattijs Jonker, and Raffaele Sommese. arXiv:2605.06880, May 2026. DOI: 10.48550/arXiv.2605.06880.
2. "A Survey of Probabilistic Micropayment Schemes." Sulyab Thottungal Valapu and Bhaskar Krishnamachari. IEEE Blockchain Technical Briefs, Technical Report, March 2022.

Software

- `tmautils`, a Python package suite of various Internet measurement utilities. Available on PyPI (package) and GitHub (source).

Teaching

- Fall 2025: Teaching Assistant, CSCI 356, Introduction to Computer Systems.
- Summer 2025: Teaching Assistant, CSCI 585, Database Systems.
- Spring 2025: Teaching Assistant, CSCI 356, Introduction to Computer Systems.
- Fall 2024: Teaching Assistant, CSCI 102, Fundamentals of Computation.
- Summer 2024: Teaching Assistant, CSCI 585, Database Systems.
- Spring 2022: Teaching Assistant, CSCI 430, Intro to Computer and Network Security.
- Fall 2021: Teaching Assistant, CSCI 270, Intro to Algorithms and the Theory of Computing.
- Spring 2021: Teaching Assistant, CSCI 102, Fundamentals of Computation.

Mentorship

- USC Viterbi Summer Institute (VSI) research mentor for incoming undergraduate students, 2022 – 2023. Students mentored: Devon Njonjo, Juan Ixcoy-Campos.

Awards

- Dean's Merit List. Indian Institute of Information Technology, Allahabad, 2017.